# General Mathematics — Exercise Sheet 19

*Matrices and composition*

**Variant: Statements**

**Exercise 1 (A matrix specified on different inputs).** A linear map $T : \mathbb{R}^2 \to \mathbb{R}^3$ satisfies

$$T(1,1) = (1,0,2), \qquad T(1,-1) = (3,2,0).$$

Find its standard matrix $A$ and justify uniqueness. Let $S : \mathbb{R}^3 \to \mathbb{R}^2$ be given by $S(x,y,z) = (x+z, y-z)$. Find its standard matrix $B$, the matrix of $S \circ T$ and the matrix of $T \circ S$. State the input and output spaces of both compositions.

**Exercise 2 (Powers with a terminating correction).** Let

$$N = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad A = I_3 + N.$$

Compute $N^2$ and $N^3$. Prove by induction that, for every integer $r \geq 0$,

$$A^r = I_3 + rN + \frac{r(r-1)}{2} N^2.$$

Compute $A^5$. Find an inverse of $A$ of the form $I_3 + cN + dN^2$, with $c, d \in \mathbb{R}$, by verifying both inverse identities.

**Exercise 3 (A rectangular complex matrix).** Let

$$C = \begin{pmatrix} \mathrm{i} & 1 & 0 \\ 0 & 1 & \mathrm{i} \end{pmatrix}.$$

Compute $C^\dagger$, $C^\dagger C$ and $CC^\dagger$, using complex conjugation from Lecture 15. A square matrix $H$ is called *Hermitian* if $H^\dagger = H$. Determine whether each of the two square products is Hermitian, and compare their traces.

**Exercise 4 (Cyclic order and an invalid interchange).** For square matrices $A, B, C$ of the same size, use Lecture 19's two-factor trace identity to show

$$\operatorname{tr}(ABC) = \operatorname{tr}(BCA) = \operatorname{tr}(CAB).$$

Determine whether one may also interchange the middle and last factors without changing the trace. Test the assertion on

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

Compute both $ABC$ and $ACB$ to justify your conclusion.

**Exercise 5 (The order of elementary operations).** Start with

$$A = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{pmatrix}.$$

First exchange rows 1 and 3. Next subtract twice the new row 1 from row 2. Finally multiply row 3 by $-1$. Find the elementary matrices $E_1$, $E_2$ and $E_3$ for these operations, their single product $E$ in the order that acts on $A$, and the final array $EA$. Describe the reverse operations that recover $A$.